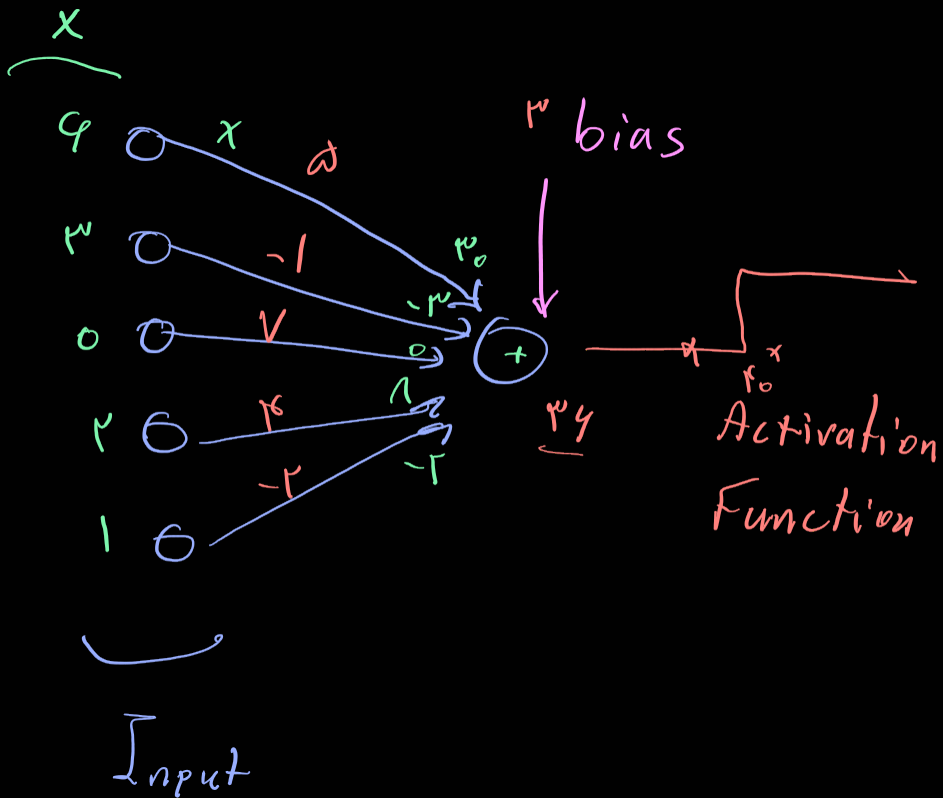
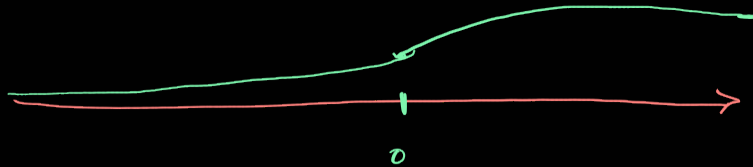
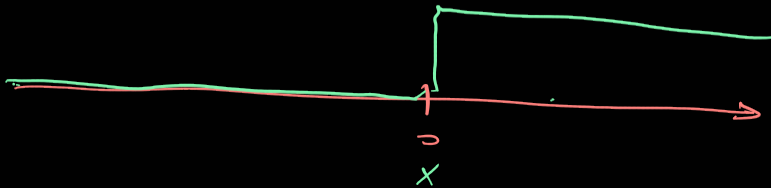




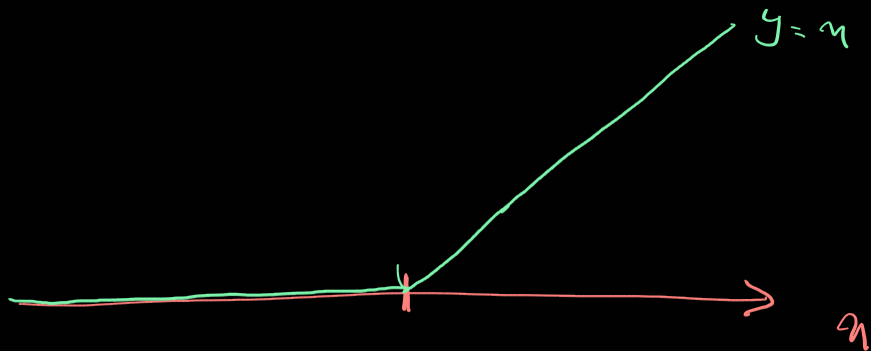
Sigmoid



$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

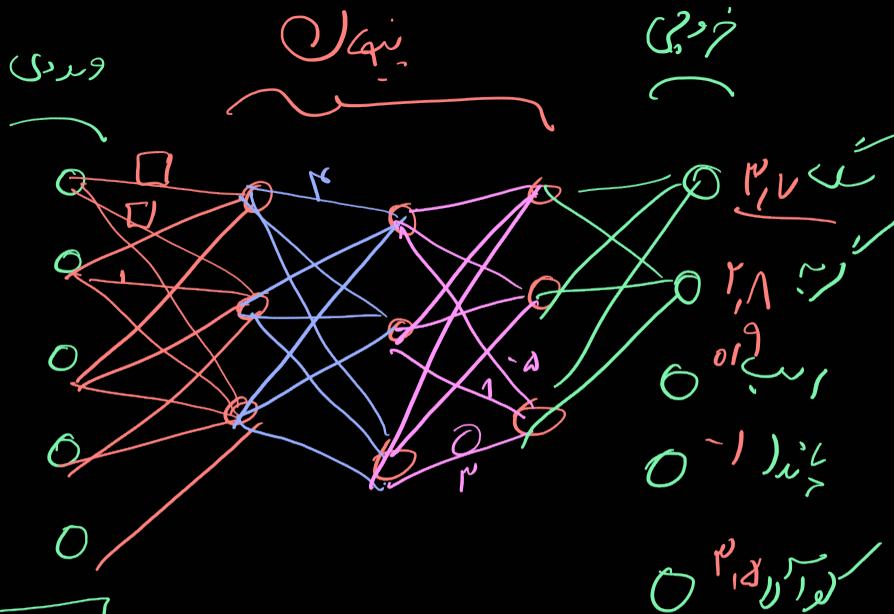


$$\text{Step}(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$



$$\text{ReLU}(x) = \max(x, 0)$$





$$\begin{aligned}
 e^{p_{1,v}} &\rightarrow \\
 e^{p_{1,\lambda}} &\rightarrow \\
 e^{p_{1,\theta}} &\rightarrow \\
 e^{-1} &\rightarrow \\
 e^{p_{1,\delta}} &\rightarrow \\
 \hline
 S
 \end{aligned}$$

Loss Function } Regression $\rightarrow$ Mean Squared Error

$$\text{MSE} = \frac{\sum_{i=1}^N (y^{(i)} - \hat{y}^{(i)})^2}{N}$$

$\rightarrow$ Mean Absolute Error

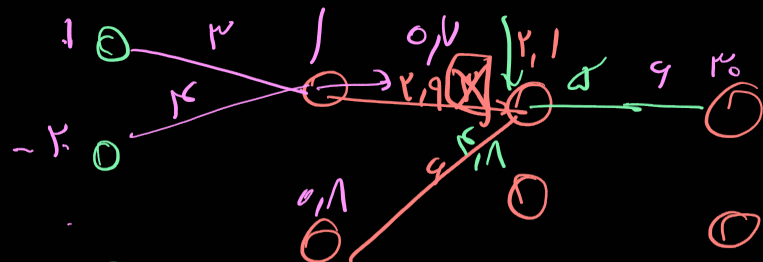
$$\text{Error} = \frac{\sum_{i=1}^N |y^{(i)} - \hat{y}^{(i)}|}{N}$$

Loss Function for

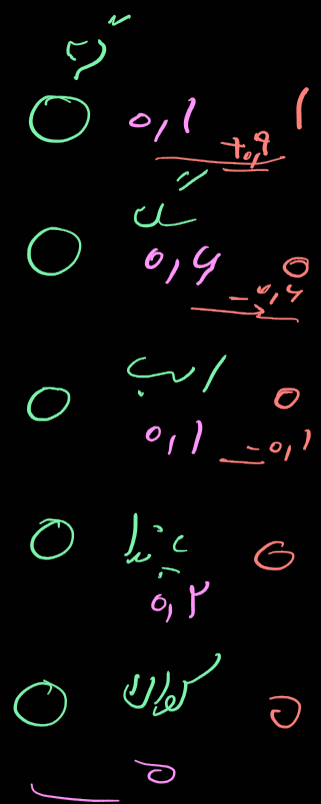
Classification

Cross Entropy

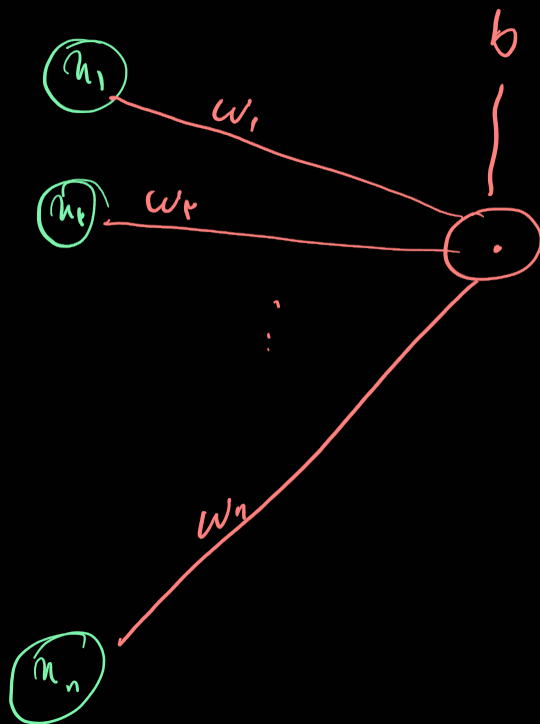
(5,2)



Feed Forward



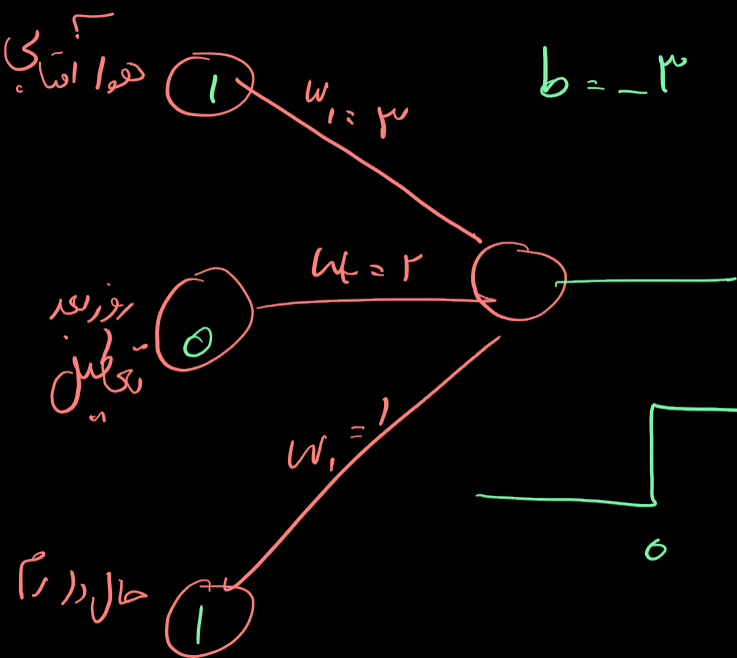
(5) 3/3



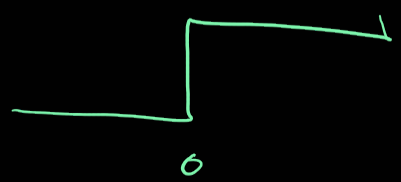
$$z = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b$$

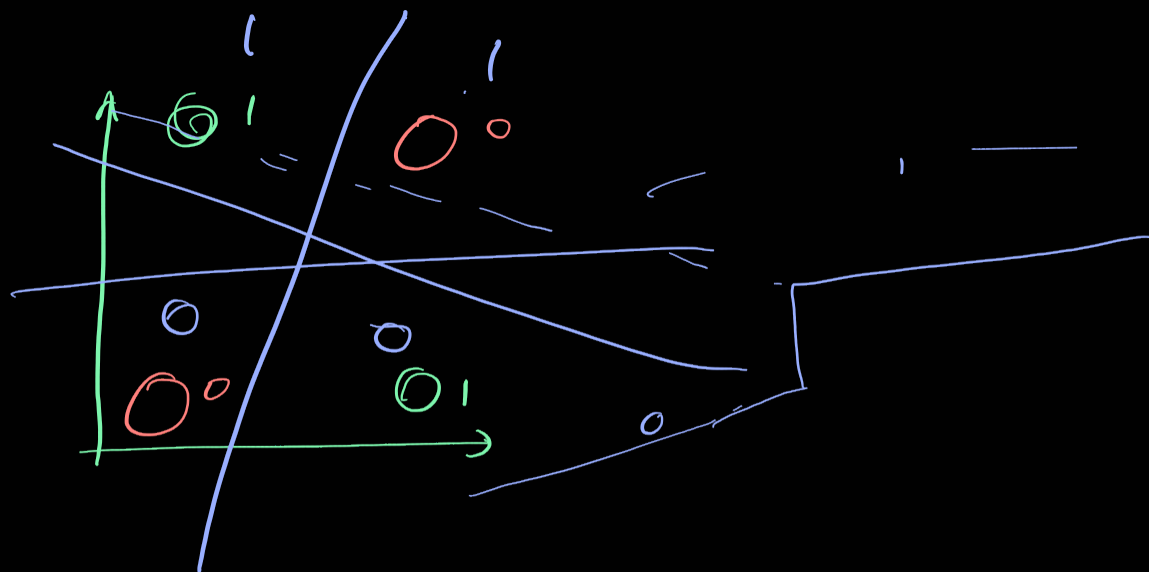
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

هوا آيا؟



$$b = 0$$





XOR

نہیں ملے گا

$$\text{Cross Entropy} = -\log(p)$$

$$\sim \log(a)$$

احتمال کلاس درست

$$a \rightarrow \begin{cases} 1 & \text{کلاس درست} \\ 0 & \text{کلاس نادرست} \end{cases}$$

$$\underline{y=1}$$

$$\underline{y=0}$$

$$\boxed{-\log(1-a)}$$

$$= \left[\underbrace{y \log a}_{\text{green}} + \underbrace{(1-y) \log(1-a)}_{\text{blue}} \right]$$

$$y=0$$